

Autosomal recessive hereditary chronic pancreatitis

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Pipeline complete — awaiting your sign-off

Repurposing hypothesis: BORTEZOMIB → Autosomal recessive hereditary chronic pancreatitis

NOTE: This candidate did NOT meet the STRONG_MATCH threshold (composite $0.4000 < 0.70$). It is included as the highest-ranked hypothesis for review; treat it accordingly.

Repurposing-only pool: the candidate compounds for this target were restricted to FDA-approved / known drugs (ChEMBL max_phase ≥ 4) at collection time. Unapproved research-grade tool compounds were excluded from the pool, not merely down-ranked.

i Pathway-neighbor candidate. The target **CTRB1** was not directly linked to **Autosomal recessive hereditary chronic pancreatitis** via Open Targets; it was discovered because it co-participates in the same Reactome pathway(s) as the primary causal gene. Open Targets association score for this target: 0.000 (0 = no direct link). The drug–target binding evidence (pChEMBL, confidence) is real; only the disease-relevance link is inferred from pathway adjacency.

▲ **Mutation-specific approval (disclosure).** BORTEZOMIB's approved / known indication explicitly names BCR-ABL. This is a DISCLOSURE flag only: it does NOT assert that the repurposing target **CTRB1** in **Autosomal recessive hereditary chronic pancreatitis** carries that mutation, and it does not change any score. The reviewer must judge whether the mutation-scoped precedent transfers to this indication.

1. Hypothesis summary

BORTEZOMIB is proposed as a repurposing candidate against **Autosomal recessive hereditary chronic pancreatitis** via the target **CTRB1**. It shows a ChEMBL median pChEMBL affinity of 6.50 at assay confidence 9/9, an Open Targets target-disease association of 0.000, and a Tanimoto similarity of 0.087 to ACARBOSE. Target network context (BioGRID, physical/genetic — not mechanism): SERPINB8, SERPINF2, DDX5, ALB, LGR4, CEP250, STK38L, KRT1. The resulting composite score is 0.4000.

Chemist rationale: Bortezomib has a median pChEMBL affinity of 6.5 against CTRB1, measured at assay confidence score 9 out of 9, and is an approved drug with a Tanimoto similarity of 0.087 to the nearest approved drug acarbose. The target CTRB1 has eight recorded BioGRID physical or genetic interactors: SERPINB8, SERPINF2, DDX5, ALB, LGR4, CEP250, STK38L, and KRT1.

STAGE 1 PRIORITIZATION SCORES

- **tractability_score:** 0.5829 (ChEMBL bioactivity + AlphaFold pLDDT – prior-trial-failure penalty)
- **unmet_need_score:** 1.0000 (treatment availability + prevalence)

These are computed by the same formulas used to rank the full rare-disease / NTD universe; a manually chosen target is scored identically, never faked or skipped.

2. Evidence table

Evidence	Value
ChEMBL median pChEMBL affinity	6.50
Assay confidence score (0-9)	9
Open Targets association score	0.000
Tanimoto to nearest approved drug	0.087 (ACARBOSE)
Approved / known drug	yes
Mutation-specific approved indication (disclosure)	⚠ YES – indication names: BCR-ABL
Lipinski/Veber (MW, logP, HBD, HBA, TPSA, rotB)	384.2, 0.36, 4, 6, 124.4, 9
Lipinski violations / Veber pass	0 / yes
PubChem XLogP (lipophilicity)	n/a
AFDB apo structure mean pLDDT (free protein, no ligand)	91.8
Boltz structure confidence (0-1)	0.946
Boltz binding-pose confidence (0-1)	0.961
Boltz predicted affinity (relative optimization score, 0-1, NOT a Kd)	0.444

Evidence	Value
Boltz predicted structure (CIF)	Download CIF (/api/structures/02a8055a687f0fe9324a2d9caff4e5c0.cif)
Boltz ADME – lipophilicity (logD)	-0.095
Boltz ADME – permeability	-0.118
Boltz ADME – solubility	high-confidence
openFDA adverse-event signal (FAERS)	PLASMA CELL MYELOMA (5209), OFF LABEL USE (3116), DRUG INEFFECTIVE (2259), NEUROPATHY PERIPHERAL (1834), THROMBOCYTOPENIA (1783)
Prior trials for this exact drug+disease	0
Target discovery method	pathway_neighbor

3. Full source citations

- **PMIDs (0):** none
- **ChEMBL activity IDs (1):** 89211
- **NCT numbers (0):** none

4. Composite score breakdown

Term	Weight	Component value	Contribution
Normalized pChEMBL affinity	0.30	0.500	0.1500
Assay confidence (score / 9)	0.20	1.000	0.2000

Term	Weight	Component value	Contribution
Normalized Open Targets association	0.20	0.000	0.0000
Normalized Tanimoto similarity	0.15	0.087	0.0130
No prior failed trial (1/0)	0.15	1	0.1500
Mechanism-direction cap (DIRECTIONALLY_INCOMPATIBLE, hard gate, max 0.400)	—	—	applied
Composite (weighted sum - penalty - cap)			0.4000

Weighted sum before penalty = 0.5131; penalty = 0.0000; reported composite_score = 0.4000. Mechanism-direction cap applied (capped at 0.400): DIRECTIONALLY_INCOMPATIBLE — The retrieved evidence states that in hereditary/chronic pancreatitis the chymotrypsin pathway is protective and the disease-relevant direction is insufficient protective activity, so inhibiting CTRB1 with bortezomib would further reduce protective trypsinogen degradation and thus worsen the core defect.

5. Limitations

- **Binding is not efficacy.** A high binding-pose confidence (0.961) or predicted affinity (0.444) only suggests the molecule may occupy the target; it does NOT establish agonism vs. antagonism, functional modulation, or therapeutic benefit.
- **ADME values are model predictions, not measurements.** The Boltz lipophilicity/permeability/solubility numbers are computed estimates and must be confirmed experimentally before any decision.

- **Structure confidence is bounded.** This hypothesis relies on a Boltz structure_confidence of 0.946 (complex pLDDT 0.937) and an AFDB apo mean pLDDT of 91.8; the AFDB model contains NO ligand, so the protein-ligand pose is entirely a Boltz prediction.
- **Assay-type and species caveats.** The affinity is a median pChEMBL over Homo sapiens IC50/Ki assays at confidence ≥ 8 ; assay heterogeneity and the bounded approved-drug reference set for Tanimoto still apply.
- **Absence of evidence is not evidence of absence.** A zero prior-trial count or no adverse-event signal may reflect that the pair has simply never been studied, not that it is safe or untried.
- **This is a repurposing *hypothesis*, not a finding.** It is a prioritised starting point that requires wet-lab and, ultimately, clinical validation.

TARGET DRUGGABILITY CONTEXT

- **No approved drug currently exists with a known mechanism against this target** (ChEMBL mechanism endpoint, Homo sapiens only).
- **Historical difficulty literature:** insufficient signal found (fewer than 2 qualifying abstracts in targeted PubMed searches for undruggability / resistance / difficulty).

Druggability context is informational only. It does not affect tractability_score, unmet_need_score, composite_score, or STRONG_MATCH.

HUMAN REVIEW CHECKPOINT

This is a falsifiable hypothesis, not a finding.

Record your scientific reasoning below, then sign off. Approving advances this candidate for wet-lab validation; rejecting closes the case permanently.

REVIEWER NOTE **required**

e.g. Affinity and structure confidence justify wet-lab follow-up despite the sub-threshold composite score.



Approve case

Reject case